DFRWS 2005 Digital Forensics Investigation Report

Tools Used:

- Autopsy
- Wireshark
- John the Ripper
- Kali Linux

Objective:

The objective of this investigation was to analyze the provided USB disk image and network captures to recover digital evidence, identify rhinoceros images, answer the forensic challenge questions, and establish any relationship between the USB evidence and the captured network traffic while maintaining forensic integrity.

Evidence Collected:

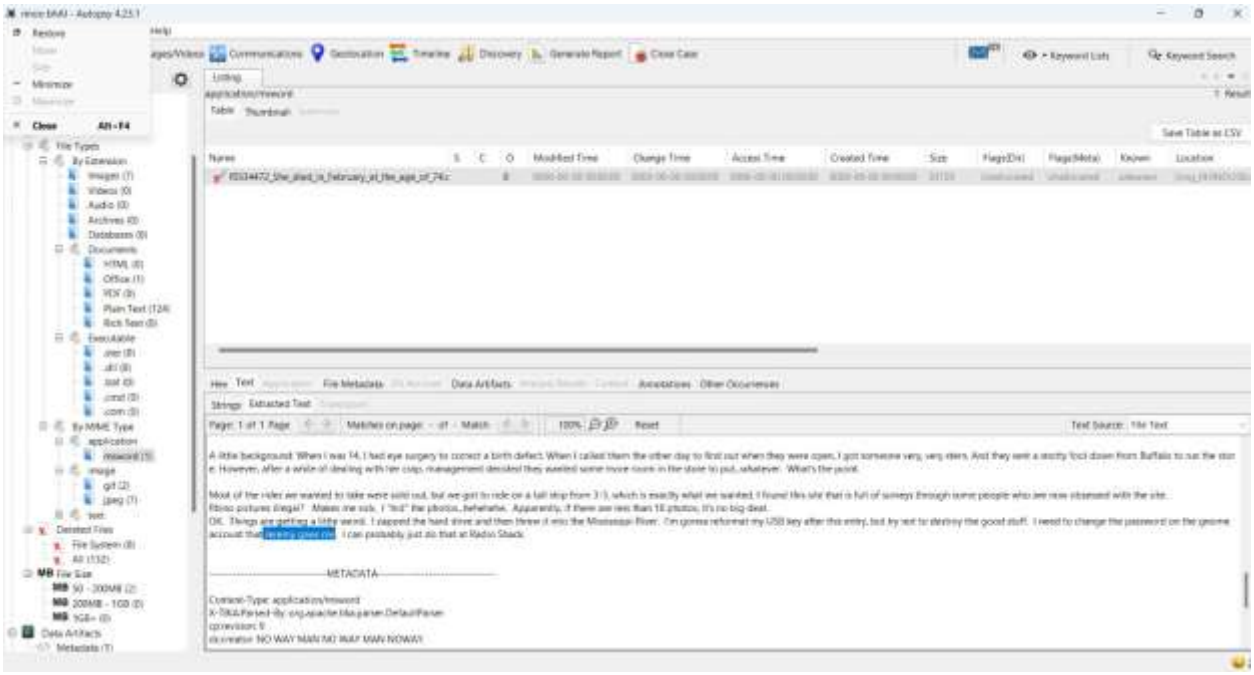
- RHINOUSB.dd (USB disk image)
- rhino.log
- rhino2.log
- rhino3.log

Questions and Answers:

Question 1: Who gave the accused a Telnet/FTP account?

Answer: Jeremy.

Evidence:



The recovered **diary.doc** contains the statement:

"...I need to change the password on the gnome account that **Jeremy gave me**."

Explanation:

The diary directly identifies Jeremy as the individual who provided the account, making this primary documentary evidence.

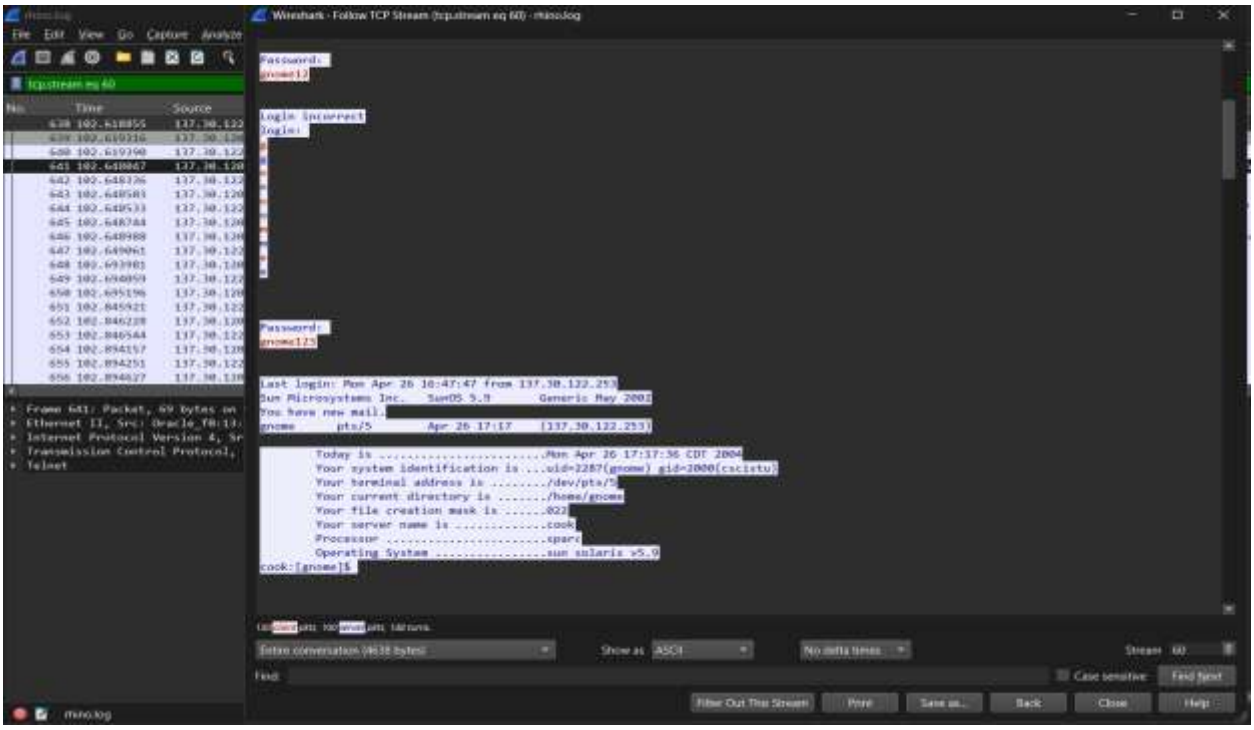
Question 2: What is the username and password for the account?

Answer:

Username: gnome

Password: gnome123

Evidence:



Recovered from the Telnet session within **rhino.log**.

The credentials appear during the Telnet authentication packets.

Explanation:

The network capture contains the login sequence in plaintext, confirming both the username and password.

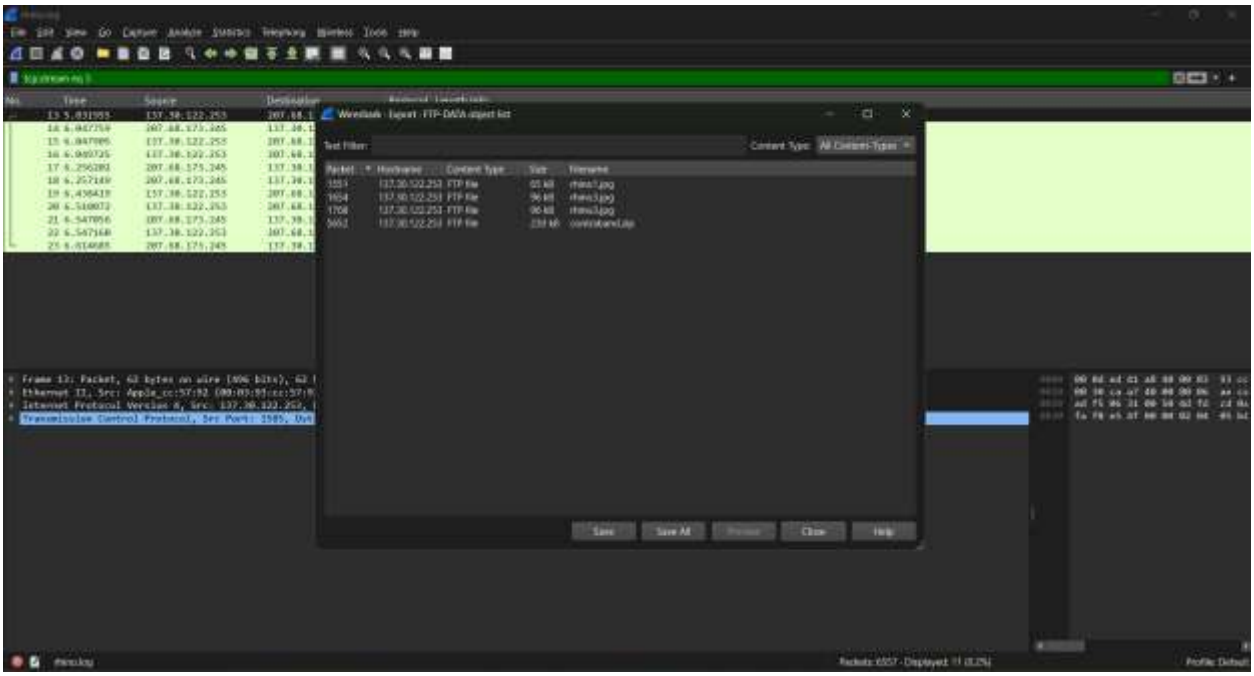
Question 3: What relevant file transfers appear in the network traces?

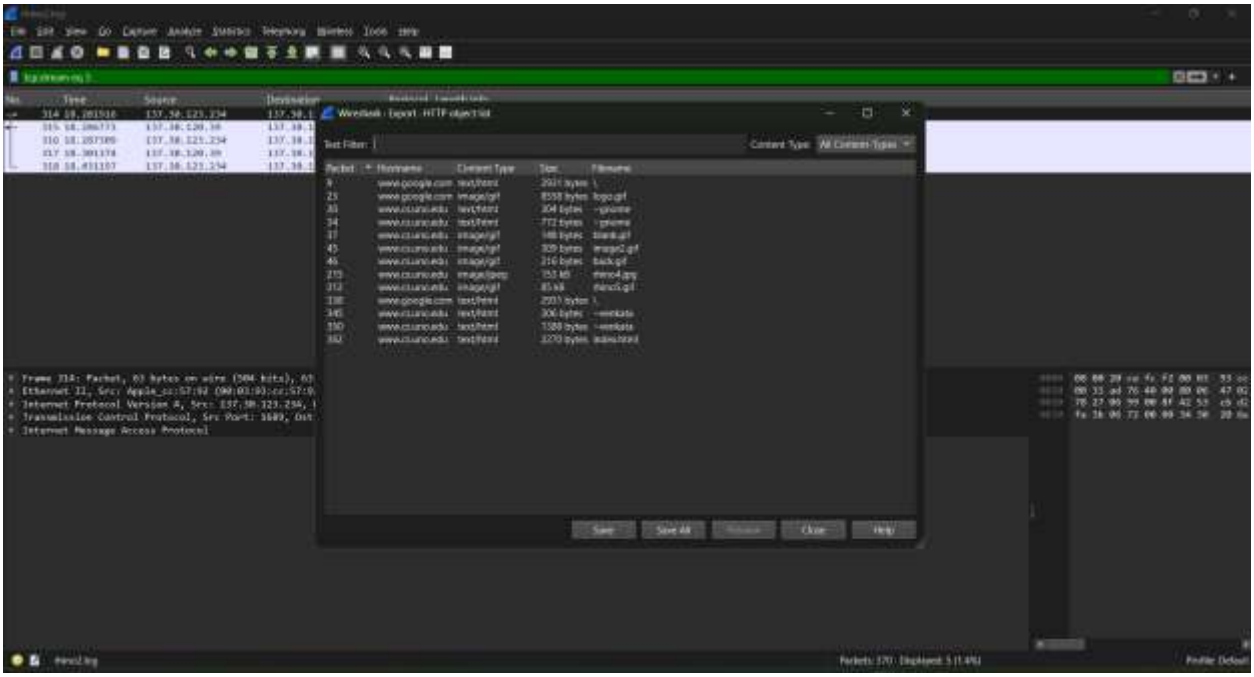
Answer

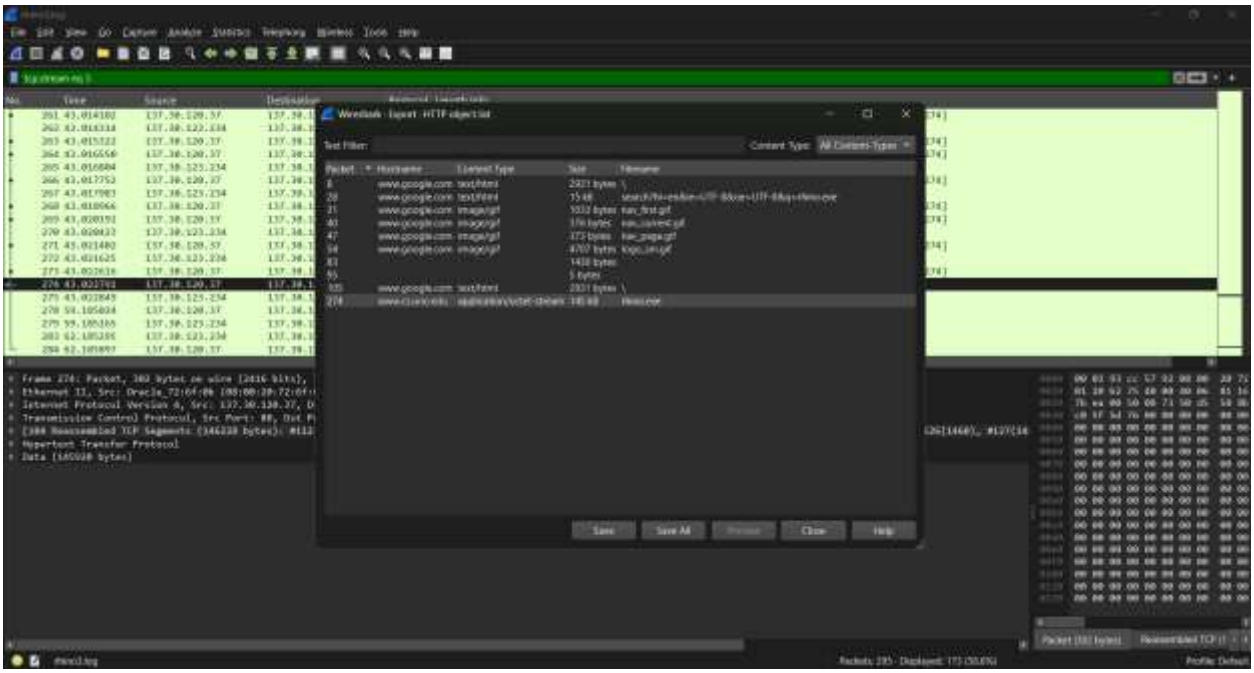
Recovered transfers include:

- rhino1.jpg (FTP)
- rhino2.jpg (inside contraband.zip)
- rhino3.jpg (FTP)
- rhino4.jpg (HTTP)
- rhino5.gif (HTTP)
- rhino.exe (HTTP)

Evidence:







Recovered from:

rhino1.jpg, rhino3.jpg in rhino.log trace [FTP transfers]

rhino2.jpg in contraband.zip file from rhino.log trace [encrypted, pw = monkey]

rhino4.jpg, rhino5.gif in rhino2.log [HTTP transfers]

An executable “rhino.exe” in rhino3.log [HTTP transfer]

Explanation:

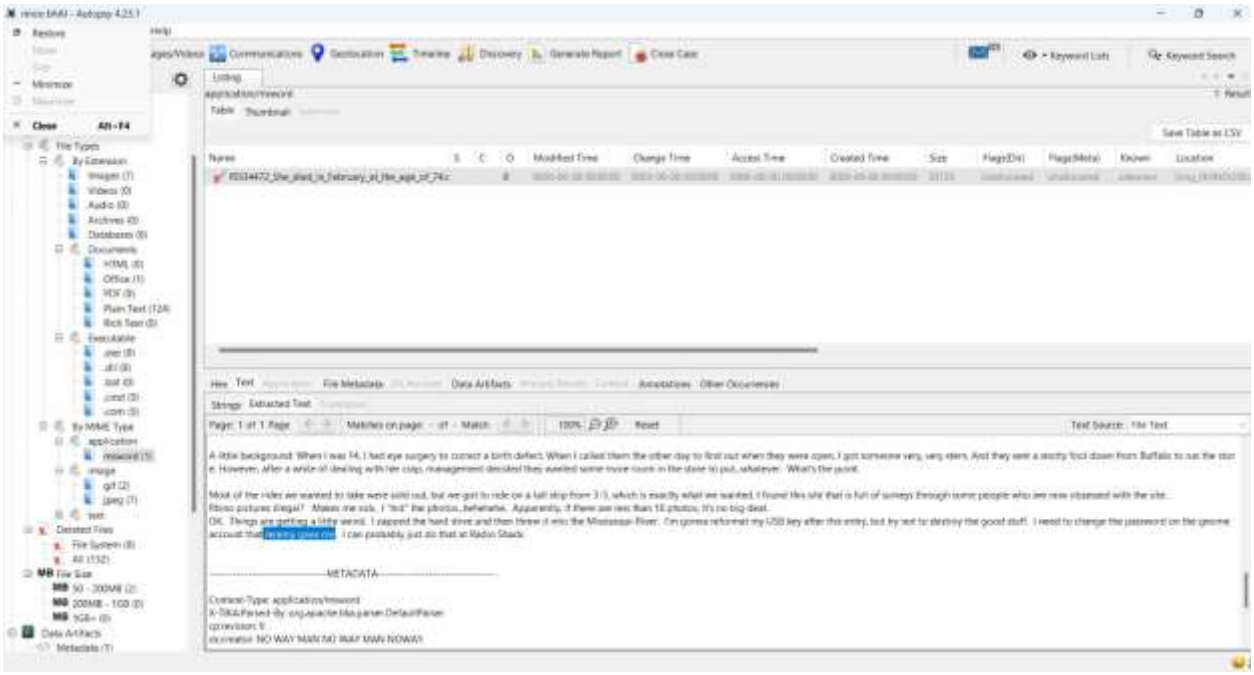
Network stream reconstruction confirms these files were transferred during FTP and HTTP sessions.

Question 4: What happened to the hard drive? Where is it now?

Answer:

The suspect erased the hard drive and threw it into the Mississippi River.

Evidence:



Recovered from **diary**:

"I zapped the hard drive and then threw it into the Mississippi River."

Explanation:

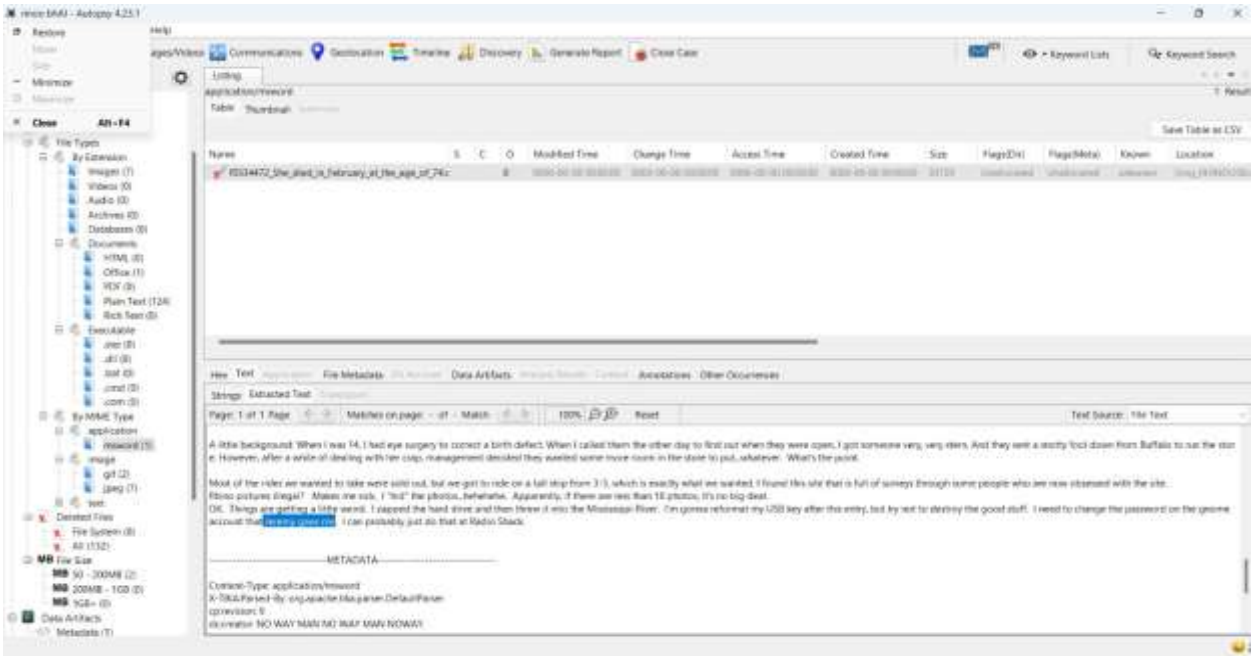
The diary provides a direct statement describing both destruction and disposal of the drive.

Question 5: What happened to the USB key?

Answer:

The USB key was reformatted.

Evidence:



Recovered from **diary**:

"I'm gonna reformat my USB key..."

Explanation:

This statement explains why deleted evidence required file carving for recovery.

Question 6:

What is recoverable from the USB disk image?

Answer:

Recovered evidence includes:

- rhino2.jpg
- rhino6.jpg
- rhino7.jpg
- rhino8.gif
- rhino9.gif
- rhino10.jpg
- diary.doc

- alligator image files

Recovered using:

- File carving
- Stegdetect
- Stegbreak
- JPSeek

Explanation:

Some files were directly carved, while others were hidden using steganography and required additional extraction.

Question 7: Is there any evidence connecting the USB image and the network captures?

Answer: Yes.

At least one thing: rhino2.jpg carved from USBKEY is same as rhino2.jpg in zip file from network trace.

Explanation:

Matching cryptographic hashes prove both files are identical, directly linking the USB evidence to the network transfer.

Conclusion:

The forensic examination successfully recovered **10 unique rhinoceros images** from the supplied evidence. Analysis of the USB image, deleted files, hidden steganographic content, encrypted archives, network captures, and diary document established a clear relationship between the storage media and the observed network activity. The recovered evidence answers all investigation questions and demonstrates that the USB device and network transfers are linked through identical recovered files and supporting documentary evidence. The findings were verified using multiple forensic techniques, including file carving, steganographic analysis, password recovery, network traffic reconstruction, and hash comparison, ensuring the conclusions are supported by reliable digital evidence.